

Assignment:- 3

Finding Grades of Students

1. Creation of text file

Wordcount.txt file will open in nano

For example -

```
student_id,subject,marks
8018,BI,85
8018,BDA,90
8018,CI,78
8018,DC,93
8028,BI,97
8028,BDA,99
8028,CI,95
8028,DC,90
8032,BI,86
8032,BDA,94
8032,CI,85
8032,DC,96
8034,BI,60
8034,BDA,55
8034,CI,40
8034,DC,40
```

Once data is added,

- press ctrl+O
- then, Enter key
- and lastly ctrl+x

to save the file and roll-back to terminal.

2. Create Mapper code file

After this command mapper_word_count.py will open in nano,
Write the mapper code for word_count

Code -

```
#!/usr/bin/env python3
import sys

# Input: student_id,subject,marks
for line in sys.stdin:
    line = line.strip()
```

```
if not line or line.startswith("student_id"):
    continue
student_id, subject, marks = line.split(",")
print(f"{student_id}\t{marks}")
```

`#!/usr/bin/env python3` - Include this line of code.

Once code is completed follow the step -

- Press ctrl + o then,
- Press Enter key,
- Lastly ctrl + x

This is save the mapper_student_grade.py and roll_back you back to terminal.

3. Create Reducer code file

After this command reducer_word_count.py will open in nano,
Write the reducer code for word_count

Code -

```
#!/usr/bin/env python3
import sys

def get_grade(avg):
    avg = float(avg)
    if avg >= 90:
        return 'A'
    elif avg >= 80:
        return 'B'
    elif avg >= 70:
        return 'C'
    elif avg >= 60:
        return 'D'
    else:
        return 'F'

current_id = None
total = 0
count = 0

for line in sys.stdin:
    line = line.strip()
    if not line:
        continue
    student_id, marks = line.split("\t")
    marks = float(marks)

    if current_id == student_id:
        total += marks
```

```

        count += 1
    else:
        if current_id:
            avg = total / count
            print(f"{current_id}\t{avg:.2f}\t{get_grade(avg)}")
            current_id = student_id
            total = marks
            count = 1

# Output for the last student
if current_id:
    avg = total / count
    print(f"{current_id}\t{avg:.2f}\t{get_grade(avg)}")

```

In above code -

`#!/usr/bin/env python3` is important to add.

Once the py code is complete

- Press ctrl + o
- Then, Enter key
- Lastly, Press ctrl + x

This will save the py file and roll-back you to terminal

If your command went wrong

Re-correct it. Before running that command make sure you delete the output create using following command -

```

hduser@Ubuntu:~$ hadoop fs -rm -r /output
2025-04-14 18:00:47,700 WARN util.NativeCodeLoader: Unable to load
native-hadoop library for your platform... using builtin-java classes
where applicable
Deleted /output

```

And then give your command -

```

hduser@Ubuntu:~$ hadoop jar/usr/local/hadoop/share/hadoop/tools/
lib/hadoop-streaming-3.3.4.jar -input /mapreduce_word_count/
word_count.txt -output /output -mapper/home/hduser/
mapper_word_count.py -reducer/home/hduser/reducer_word_count.py

```

Or else you will get following error -

```

2025-04-14 18:00:19,901 WARN util.NativeCodeLoader: Unable to load
native-hadoop library for your platform... using builtin-java classes
where applicable
2025-04-14 18:00:20,774 INFO impl.MetricsConfig: Loaded properties
from hadoop-metrics2.properties

```

2025-04-14 18:00:21,121 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).

2025-04-14 18:00:21,122 INFO impl.MetricsSystemImpl: JobTracker metrics system started

2025-04-14 18:00:21,163 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!

2025-04-14 18:00:21,374 ERROR streaming.StreamJob: Error Launching job : Output directory hdfs://localhost:54310/output already existsStreaming Command Failed!